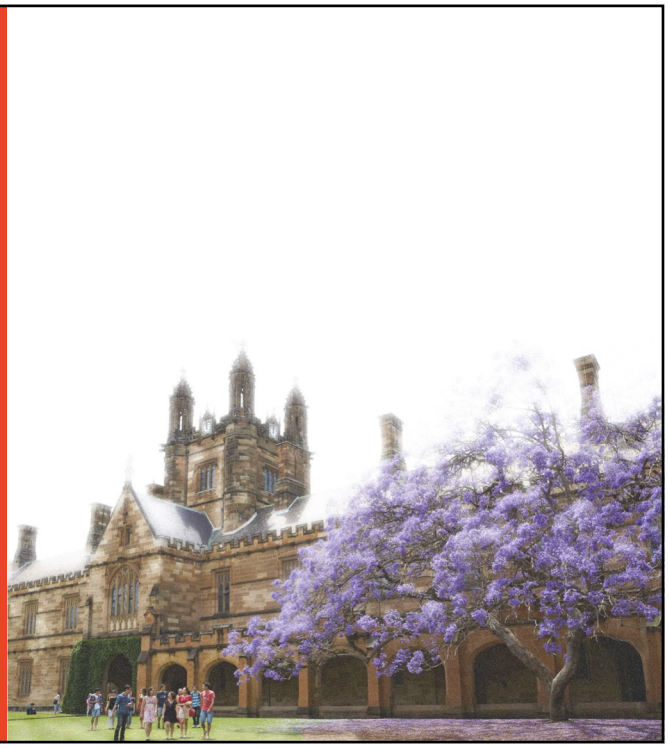
Detecting Content Rating Violations in Android Applications: A Vision-Language Approach

Suranga Seneviratne

TrustCom'25 - IEEE International Conference on Trust, Security and Privacy in Computing and Communications

THE UNIVERSITY OF SYDNEY



1



Video Game Content Ratings: Does Anyone Care Anymore?

the Artifice

When video game age ratings go wrong: Balatro's battle with Pegi

Card game Balatro was a huge hit last year - but a year-long struggle with the European age ratings board has exposed how well-meaning rules can have unintended consequences



Kids exposed: Video game age ratings still failing Australian families

Mislabelled apps are exposing children to gambling-like play

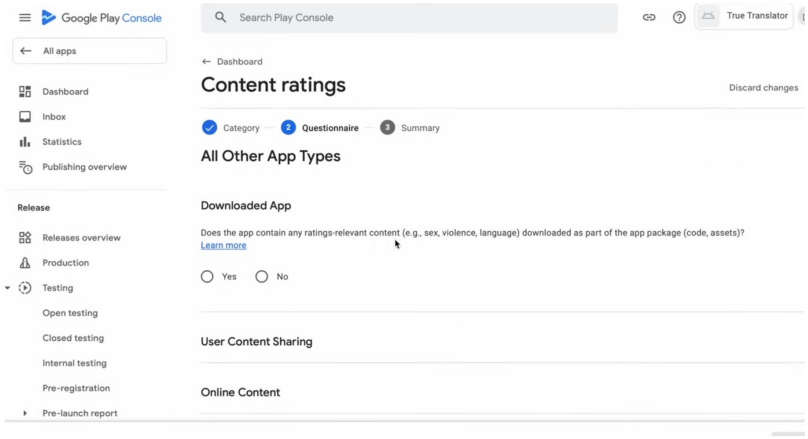
2

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Google Play Store: How to Determine the Content Rating?

App Developer

Fill a Questionnaire based on the content



Google Play Store: How to Determine the Content Rating?

App Developer

Fill a Questionnaire based on the content

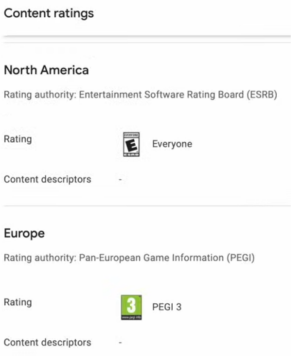
IARC
INTERNATIONAL AGE RATING COALITION

Determines the Content Rating

Google Play

Previews How Regional Content Ratings Look Like





Google Play Store: How to Determine the Content Rating?



5

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Malpractices, Violations & Loopholes



6

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Literature

Automatic App Maturity (content) Rating Prediction:

- Text only: [Chen et al. 2013]: Text-mining with keyword matching; lacks semantic understanding.
- Text only: [Hu et al. 2015]: Uses text features with SVM based online training.
- Multimodal: [Liu et al. 2016], [Zhou et al. 2022]: Combine image and APK data but treat modalities separately.
- Gap: Existing studies rely on single-modality or shallow and independent multimodal analysis.

Vision–Language (VL) Models

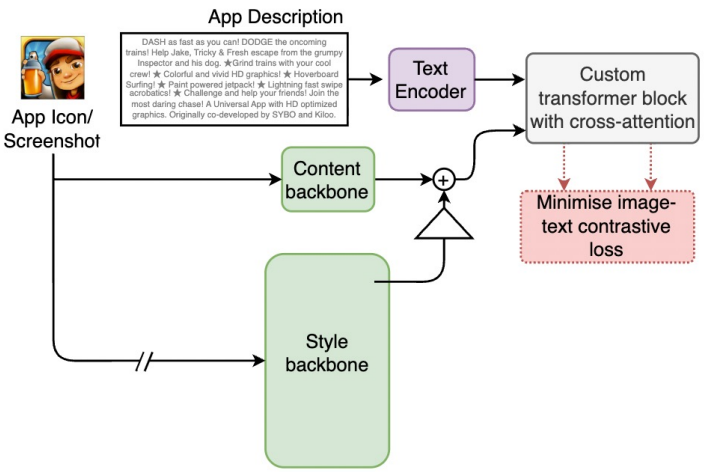
- Contrastive models CLIP, ALIGN [2021]: Align image–text embeddings via symmetric cross-entropy loss.
- Learned embeddings reused for various vision–language tasks
  - [Portillo-Quintero et al. 2021], [Nichol et al. 2021], [Massiceti et al. 2023]
- LMM adaptation for domain-specific tasks is challenging, especially on different domains than pre-trained.
  - [Agarwal et al. 2021], [Luccioni et al. 2024]
- This work: First to apply advanced VL models for app "*content compliance*" detection.

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Our Framework

(a) INTUITION



- Why text?
- Why image content?


Example app icon
From [G] rated game


Example app icon
From [M] rated game

- Why image style?



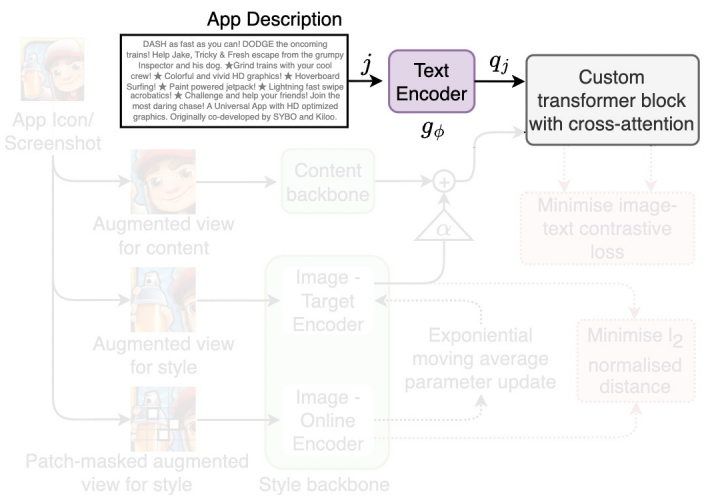


8

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Our Framework

(b) TEXT ENCODER



App Description

- Max allowed 4,000 characters
- Average 2,169 characters

RoBERTa (110M) Encoder

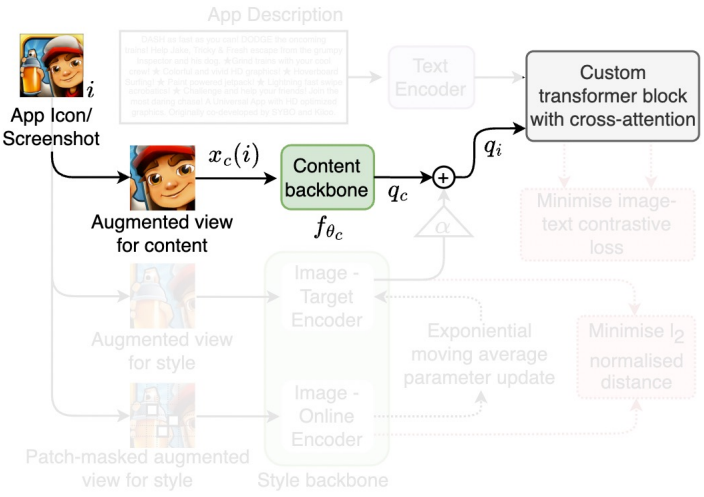
- 256 max tokens
- Randomised text chunking
- Four or more consecutive sentences randomly extracted from long app descriptions

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Our Framework

(c) CONTENT BACKBONE



CLIP image encoder (XX parameters)

- Always active in training

Operates on augmented images of 224x224 resolution.

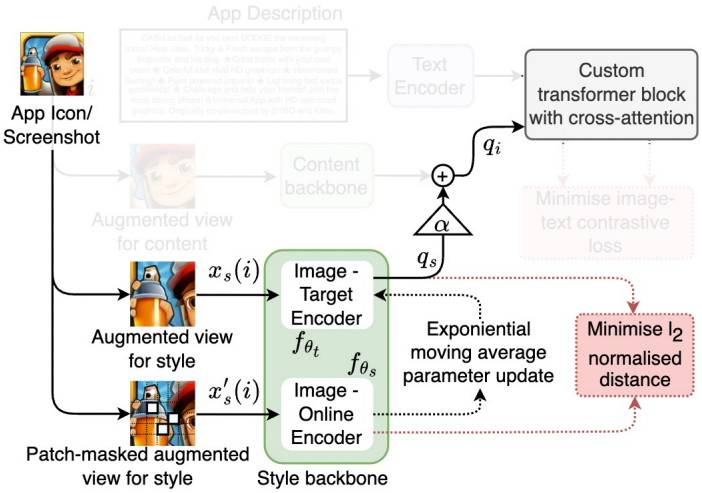
- We follow the augmentation settings defined in CLIP to generate different views of the image.

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Our Framework

(d) STYLE BACKBONE



CLIP image encoder (XX parameters)

- Masked representation learning task

Operates too on augmented images;

- Online Network:
  - randomly mask three 3 × 3 patches to produce the masked image
- Target Network:
  - Unmasked image passed through
  - A slow-moving average network

Training: EMA for stability

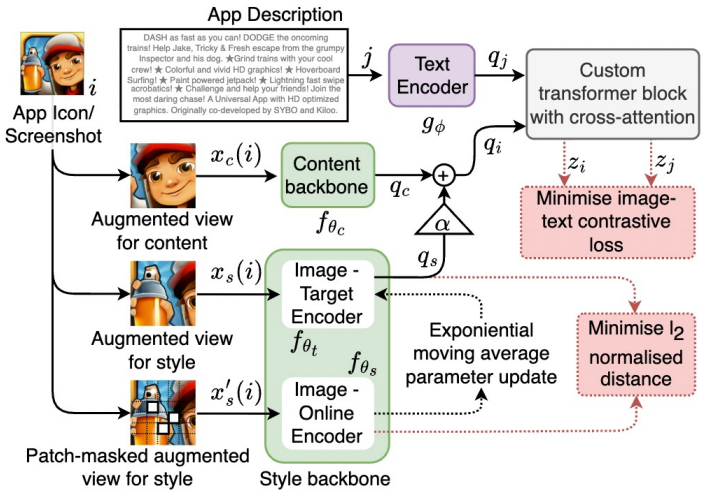
- $\theta_t \leftarrow \tau\theta_t + (1-\tau)\theta_s$

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Our Framework

(e) HOW TO MERGE?



1) Text description may not perfectly reflect what is depicted in the app icon or screenshots

2) Some modalities may not always contain useful information related to the content/age rating

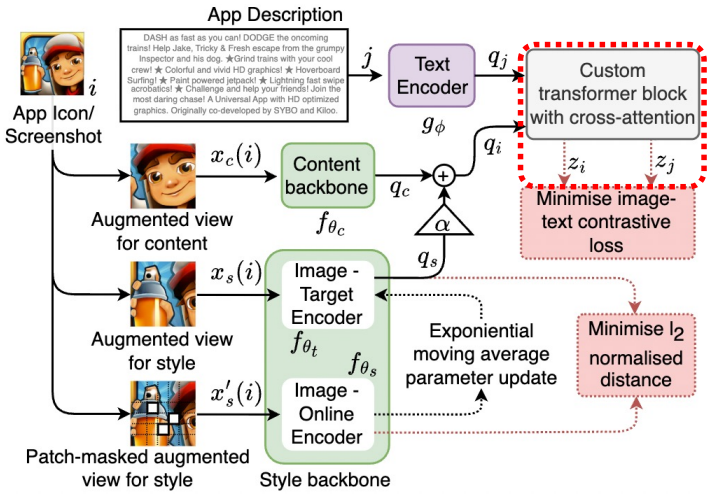
3) App images display complex variations within the same content rating class

12

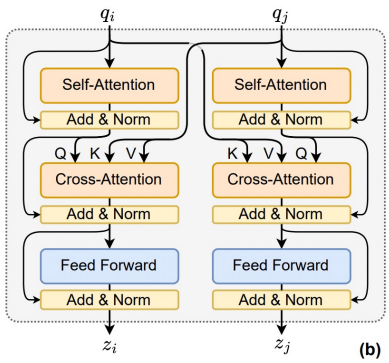
12

Our Framework

(f) CUSTOM TRANSFORMER BLOCK



Self-attention followed by cross-attention



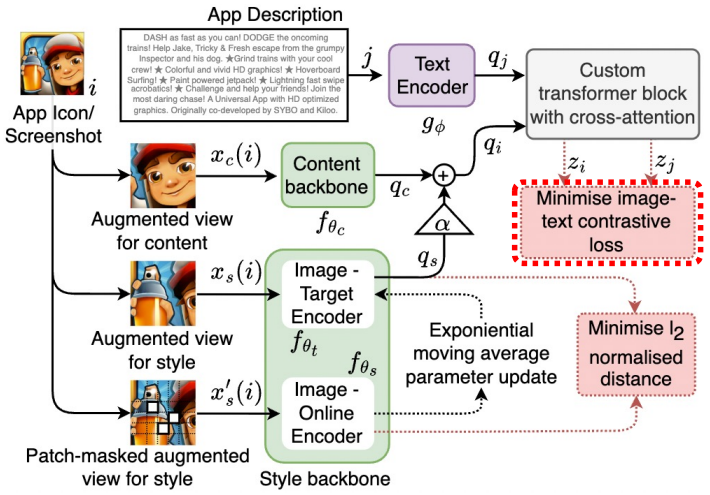
Linear projection layer outputs the final visual and textual embeddings.

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Our Framework

(g) OUR LOSS FUNCTION – PART I



$$L = L_{SigCL} + \lambda L_{mse}$$

1. supervised Contrastive Loss, more specifically, Sigmoid Contrastive loss (SigCL) with content rating labels

Positive pairs = image-text pairs with the same rating

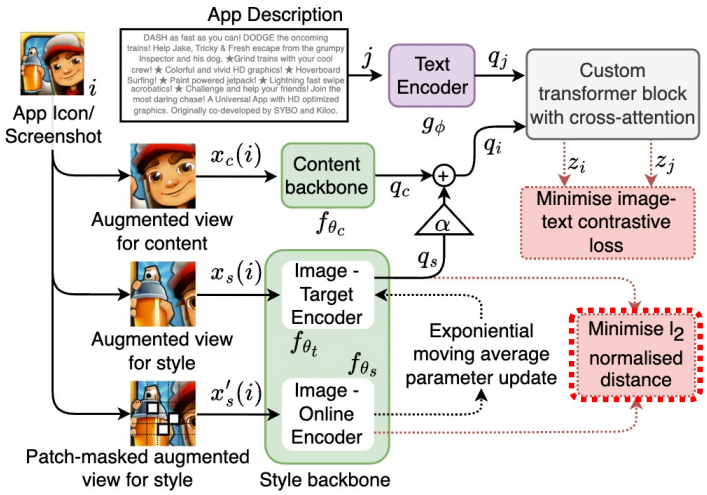
$$L_{SigCL} = -\frac{1}{|P|}\sum_{i,j\in P}\log\frac{1}{1+e^{(-\tau z_i \cdot z_j + b)}} \quad -\frac{1}{|B|}\sum_{i\in|B|}\sum_{j\in|B|\setminus\{P\}}\log\frac{1}{1+e^{(\tau z_i \cdot z_j - b)}}$$

14

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Our Framework

(h) OUR LOSS FUNCTION – PART II



$$L = L_{SigCL} + \lambda L_{mse}$$

2. Lmse optimises two views encoded by the target and online networks

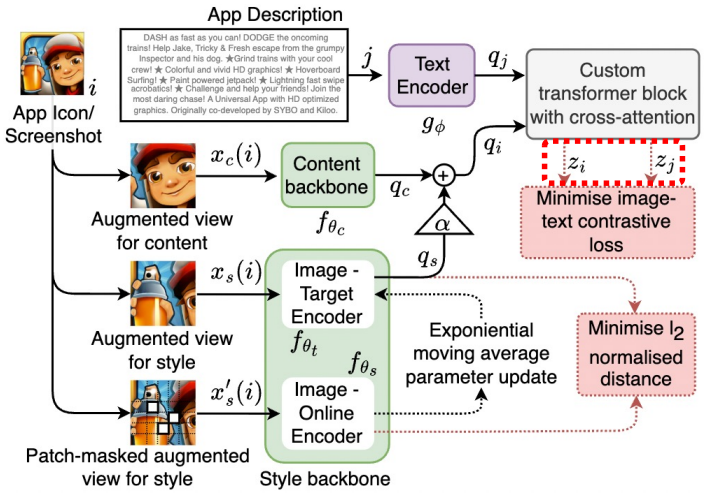
$$L_{mse} = \|q_s - q'_s\|_2^2$$

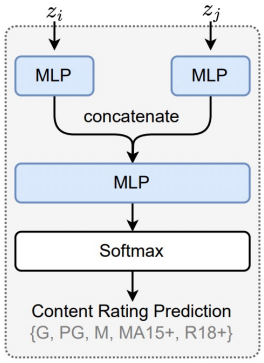
15

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Our Framework

(i) DOWNSTREAM TASK – CONTENT RATING PREDICTION





	Train	Valid.	Test	Teacher Approved
G	4,544	1,139	2,650	2140
PG	4,540	1,130	2,649	30
M	4,530	1,120	2,648	2
MA15+	2,131	547	1,796	-
R18+	255	62	255	-

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Results

	Macro Average		Weighted Average		
	Precision	Recall	Precision	Recall	Accuracy
ViT + RoBERTa	33.33	33.87	48.45	50.78	50.78
BLIP	33.26	32.59	47.73	49.13	49.13
CLIP	53.66	45.44	49.97	50.11	50.12
CLIP fine tuned	57.70	45.56	50.60	50.19	50.18
Ours	61.57	46.63	54.80	53.09	53.09

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Ablation Studies

	a Different Loss Types			b Different Structural Formats		
	SCE	UniCL	Ours*	Without Style encoder	Without cross attention	Ours*
Micro Precision	56.36	60.29	61.57	62.60	58.81	61.57
Micro Recall	38.52	43.45	46.63	43.56	46.43	46.63
Accuracy	48.59	51.52	53.09	50.73	50.76	53.09

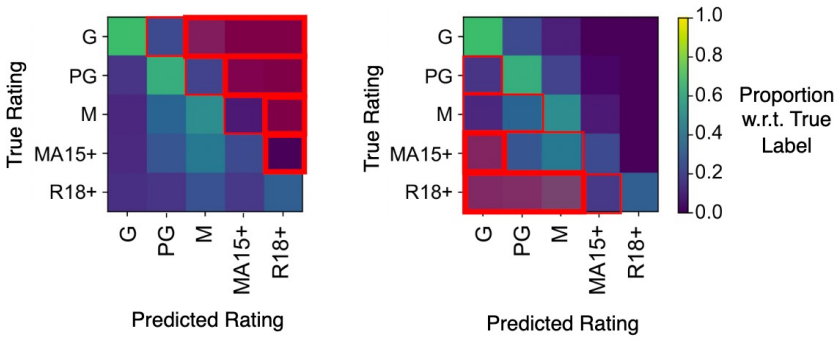
	c Image Embeddings Only			d Text Embeddings Only			Ours*
	ViT	CLIP ft	Ours	RoBERTa	CLIP ft	Ours	
Micro Precision	28.33	56.64	61.18	52.35	49.25	60.58	**61.57**
Micro Recall	35.25	42.39	45.14	46.86	41.63	43.31	**46.63**
Accuracy	46.07	49.79	51.15	49.84	50.18	50.46	**53.09**

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Predictions in the Wild

(a) TEST SET OF ~10,000 GAMES



Pred > True

Potential Malpractices

Pred < True

Potential Disguises

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Predictions in the Wild

(b) POTENTIAL MALPRACTICES – HOW DO THEY LOOK?

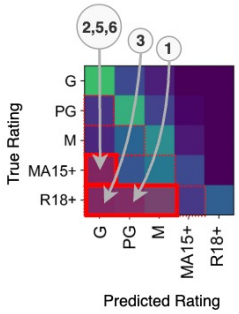


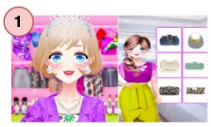
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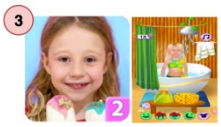
Predictions in the Wild

(c) POTENTIAL DISGUISES – HOW DO THEY LOOK?

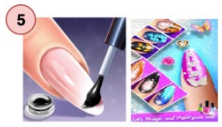




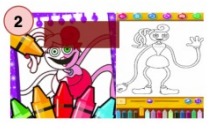
Pred: [PG]
True: [R18+]



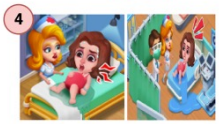
Pred: [G]
True: [R18+]



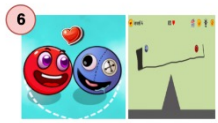
Pred: [G]
True: [MA15+]



Pred: [G]
True: [MA15+]



Pred: [G]
True: [M]



Pred: [G]
True: [MA15+]

Summary

- Multi-Modal Vision-Language Model:
  - A custom model combining image and text encoders with cross-attention to predict app content ratings from creatives and descriptions.
- Performance:
  - Achieved 5.9% and 5.8% relative accuracy improvements over CLIP and fine-tuned CLIP models on a large dataset of gaming apps.
- Real-World Impact:
  - Authors Identified 71 apps (~17%) with potential rating malpractices;
  - Notably, 16.86% of these were later removed from the Play Store.
- Behavioural Insights:
  - Detected 32 apps correctly rated but visually misleading—appearing child friendly.
- Future Directions:
  - Enhance the model by integrating user feedback, data safety declarations, and dynamic behaviour signals to improve large-scale detection and regulatory support.

https://arxiv.org/abs/2502.15739

Contact Us:
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Thank You

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